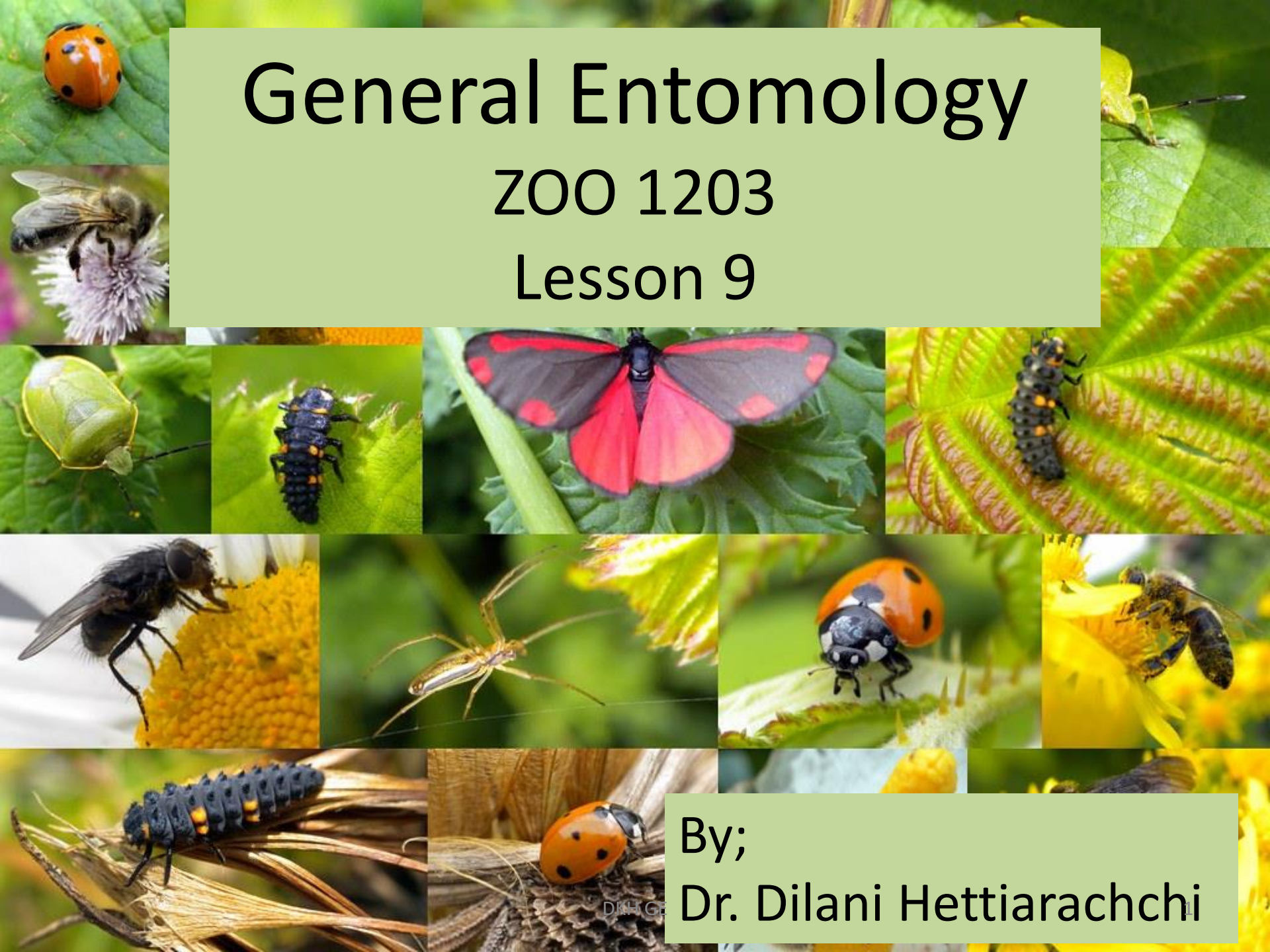


General Entomology

ZOO 1203

Lesson 9



By;
Dr. Dilani Hettiarachchi

Lesson Objectives

After completion you should be able to;

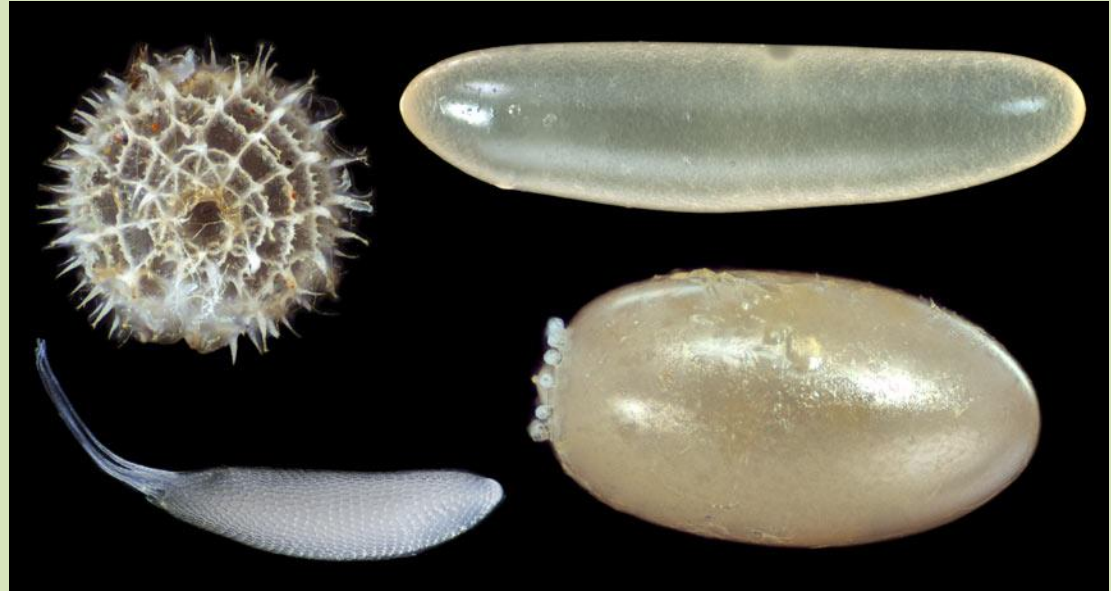
- Describe the stages of reproduction in insects
- Explain the stages of insect life cycle
- Describe the process of metamorphosis
- Discuss different pupal and larval forms seen in the insect world
- Explain the link between seasonal cycles and life cycle when controlling insects

Development of Insects

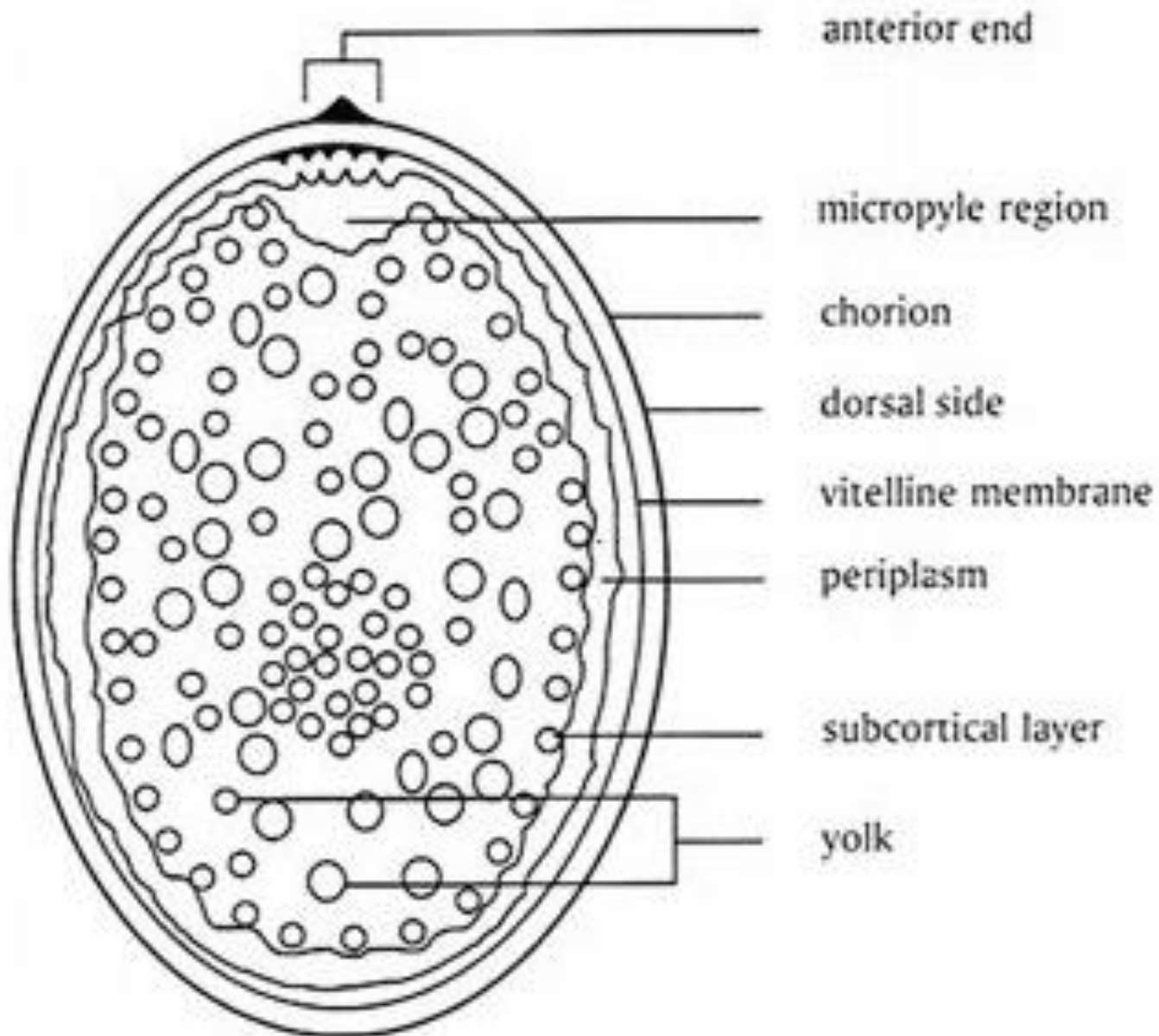
- Chain or sequence of biological events that occurs during the lifetime of an individual insect is called “Insect Life Cycle”
- Deposition of eggs is the starting point
- known as **ovipary**
- Egg is released from female body via an ovipositor
- Production of eggs → oogenesis
- Egg laying → oviposition



- Eggs are genetically unique and physically distinctive



- Regardless of size or shape each egg composed of only a single living cell → female gamete
- egg is fertilized by sperms from the spermatheca
- Sperm nucleus fuse with the egg nucleus

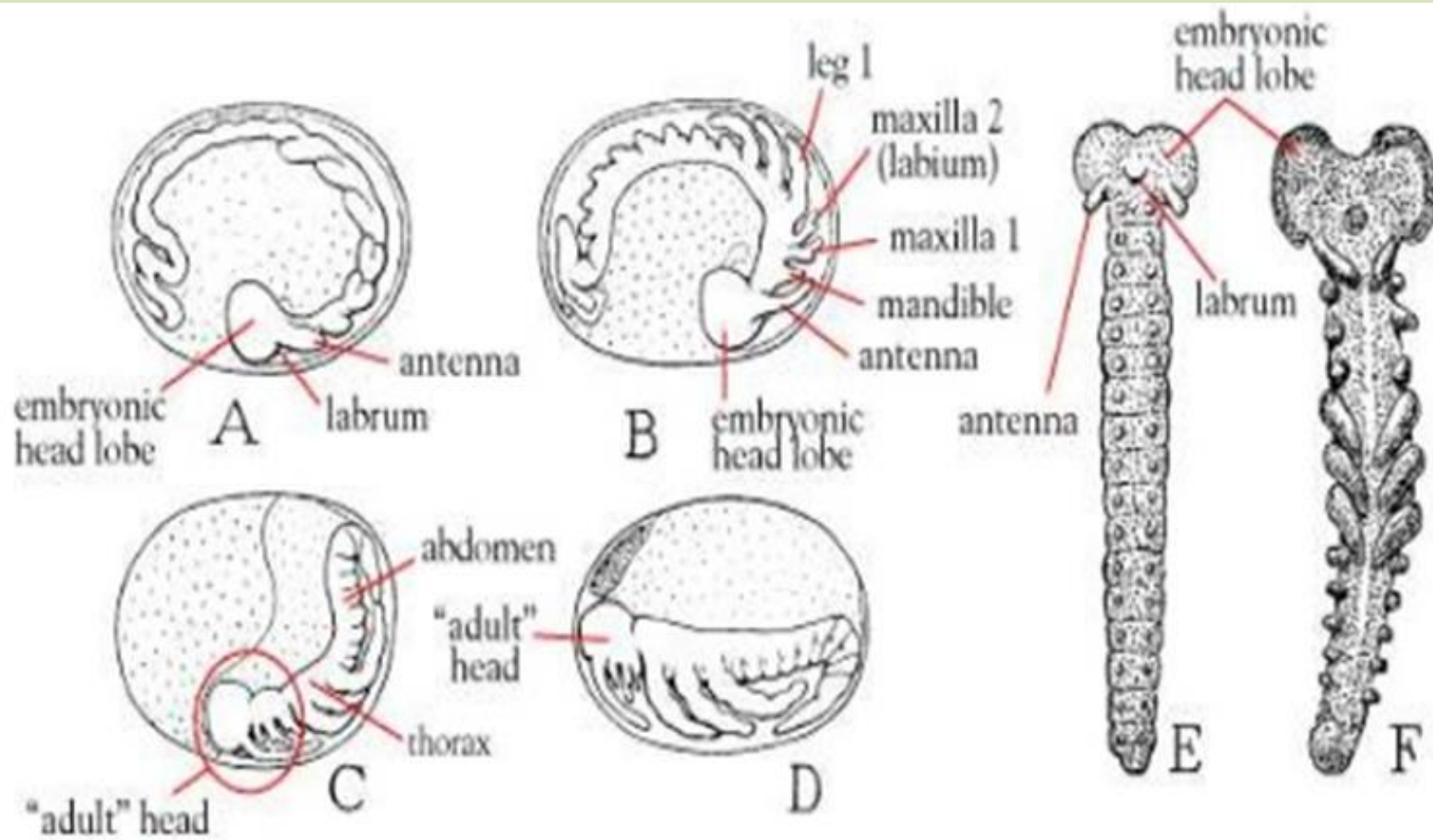


- After the egg is fertilized, it undergoes a period of rapid growth and development known as embryogenesis, literally the "embryo's beginning"
- During early development:
 - embryo's body is worm like
 - there three germ layers will form:

Ectoderm → epidermis, exocrine glands, brain and nervous system, sense organs, foregut and hind gut, respiratory system, external genitalia

Mesoderm → heart, blood, circulatory system, muscles, endocrine glands, fat body, gonads

Endoderm → midgut



A-D, Four stages in the development of an earwig inside the egg, lateral view.

E, Diagrammatic ventral view of early insect embryo. F, Ventral view of mantis embryo at a later stage in development.

Bud like swelling in front of the mouth opening and grow to form labrum and antennae

ment first become the anterior end

Embryonic Development

Appendages

Mandibles

Maxillae

Labium

Abdomen

Walking legs

Development time varies from species to species

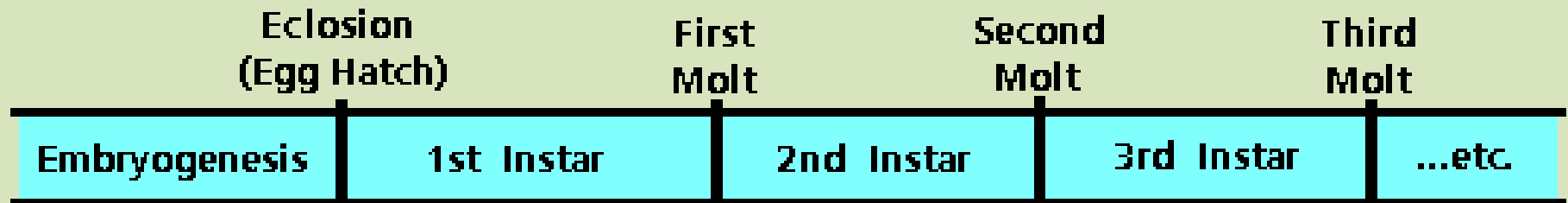
Day 5

Day 7

- Rate of embryonic development depends on :
 - Temperature
 - Species-specific characters of development
- Yolk's contents have been consumed → embryogenesis ends
- Hatching process → eclosion
 - Young insect may chew its way through the egg's chorion
 - Or it may swell in size by imbibing air until the egg shell cracks
- Called the first instar nymph/larva

- First instar → small, wingless, sexually immature
- Primary role → eat and grow
- Undergo molting
- Collectively all changes that involved growth, molting and maturation →

Timeline of Morphogenesis



Polyembryony

- Some more than one embryo forms in a single cell through a process of asexual division (eg: parasitic Hymenoptera)
- Called POLYEMBRYONY
- No of embryos forms can vary – from 2-3 to 100 -3000
- each embryo develops into larvae
- Advantage to parasites – single host one egg large no. of larvae

Reproductive strategies

- Sexual reproduction
- Asexual reproduction or Parthenogenesis
eg: several black flies (Diptera: Simuliidae)
number of weevils (Coleoptera: Curculionidae)
Aphids (Homoptera: Aphididae)
- A combination of sexual and asexual modes –
haplodiploidy/ Facultative parthenogenesis
 - Eg: Coleoptera Hymenoptera
 Thysanoptera Hemiptera

OVIPAROUS

- Most are oviparous
- Embryonic development proceed within the egg after it has been laid/deposited

OVOVIVIPAROUS

eg: many flesh flies (Diptera)

- Egg hatch when these eggs are still in the female egg-laying organ

VIVIPAROUS

eg: *Glossina* sp./tsetse fly

- - One larvae develops at a time in a specialized “uterus”

Metamorphosis

- Development process that takes place from eclosion of the free-living insect until adulthood is called metamorphosis
- “change in form”
- Changing from a worm-like crawling form to an ovoid inactive form and finally to a winged flying form
- Or there might not be visible changes except for the size

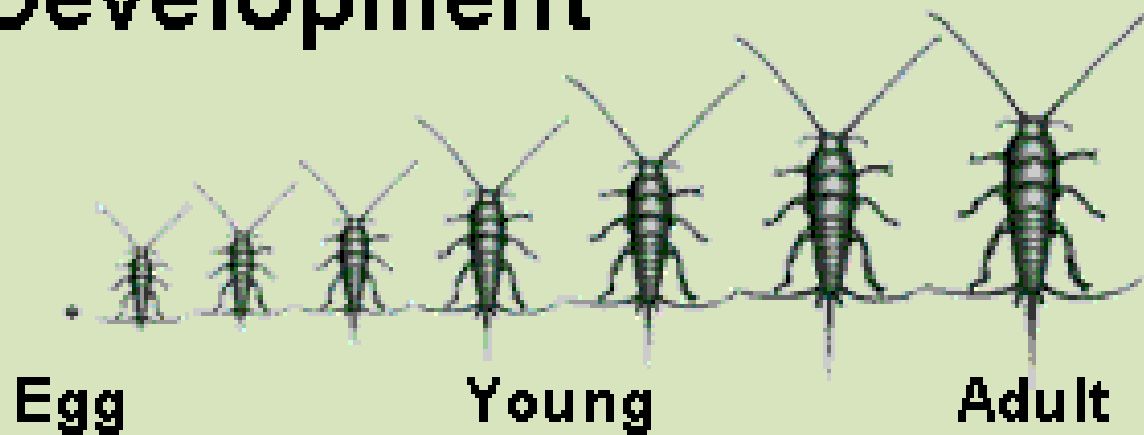
- Metamorphosis represents orderly, genetically programmed changes in insect form
- Radical changes results due to rearrangements with in cells, tissues and organs
- Two phases; Histolysis and Histogenesis
- Primarily under the control of juvenile hormones (JH)
- Development and growth are interconnected



Ametabolous Development

- insects undergo little or no structural changes as they grow
- Immatures are called young
- the young resemble the adults in all respects except in size and sexual structures.
- It grows only in size by molting
- Eg: silverfish

Ametabolous Development



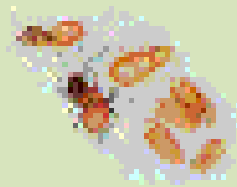
Hemimetabolous Development/ incomplete metamorphosis

- Exhibit gradual changes in body form during morphogenesis
- Immature called nymphs or if aquatic naiads
- Have external wing pad
- special tracheal gills → aquatic respiration
- Spend most of their time in different habitats (adults and larvae)

Hemimetabolous Development



Eggs



Nymphs



Adult

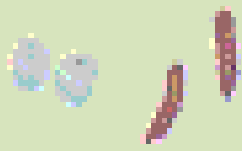
Holometabolous Development/ complete metamorphosis

- Immature → larvae
- In case of beetle (Order Coleoptera) the larva is known as grub
- Larvae are very different from adults
- Larvae are “feeding machines”
- undergo dramatic changes in ecology, physiology and morphology at the larval to adult transition

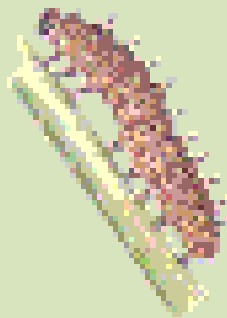
- become larger at each molt but do not acquire any adult-like characteristics
- When fully grown, larvae molt to an immobile pupal stage and undergo a complete transformation
- The adult stage, which usually bears wings, is mainly adapted for dispersal and reproduction
- Ex: Beetles, Flies, Mosquitoes, Moths and Butterflies

Holometabolous Development

Eggs



Larvae



Pupa



Adult



Most larvae can be grouped into five categories based on physical appearance

1. Eruciform

Common name: Caterpillar

Description:

Body cylindrical, short thoracic legs, 2 – 10 pairs of fleshy abdominal prolegs

Example: moths and butterfly



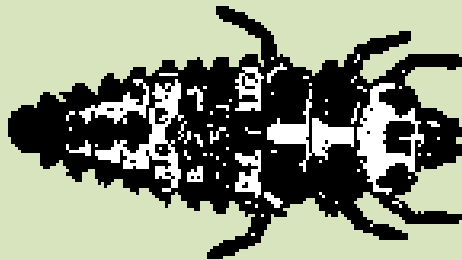
2. Campodeiform

Common name: Crawler

Description:

Body elongated, flattened, prominent antennae and/or cerci, thoracic legs adapted for running

Example: lady beetle, lacewing



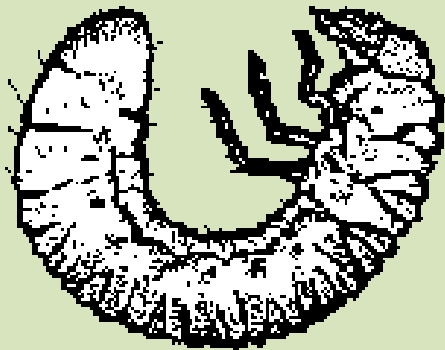
3. Scarabaeiform

Common name: White grub

Description:

Body robust and “C” shaped, no abdominal prolegs, short thoracic legs

Example: dung beetle, June beetle



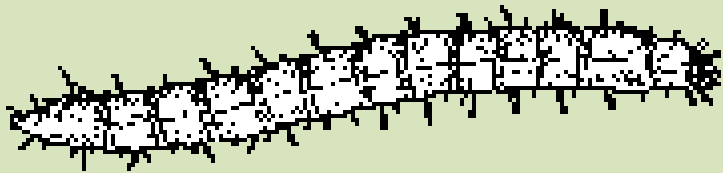
4. Elateriform

Common name: Wireworm

Description:

Body long, smooth, cylindrical with hard exoskeleton, very short thoracic legs

Example: flour beetle, click beetle



5. Vermiform

Common name: Maggot

Description:

Body fleshy, worm-like, no head capsule or walking legs

Example: house fly, flesh fly



Pupae can be grouped into three categories based on physical appearance

1. Obtect

Common name: Chrysalis

Description:

Developing appendages (antennae, wings, legs, etc.) held tightly against the body by a shell-like casing. Often found enclosed within a silken cocoon

Example: Butterflies and moths



2. Exarate

Description:

All developing appendages free and visible externally

Example: Beetles, Lacewings



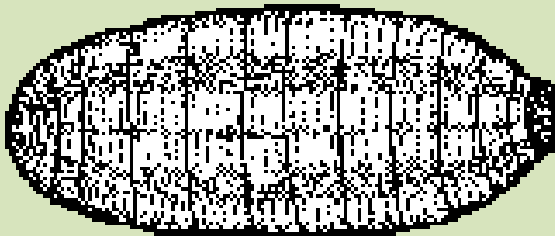
3. Coarctate

Common name: Puparium

Description:

Body encased within the hard exoskeleton of the next-to-last larval instar

Example: Flies



Periodic behaviour

- exhibit cycles of activity that are synchronized with periodic events in their environment
- When these periods of activity are repeated at about the same time each day, an animal is said to exhibit **diel periodicity**

Diurnal insects are active during daylight hours

Nocturnal insects are active at night

Crepuscular insects are active at dawn and dusk

Matinal behavior occurs only near dawn

Vespertine behavior occurs only near dusk

- In addition to daily rhythms, many insects also exhibit periodic behaviors with much longer cycles

Circalunar rhythms are synchronized with the phases of the moon and have a cycle of about 28 days. Many insect species tend to molt or emerge as adults during the dark phase of the moon as an adaptation to evade predators.

Circannual rhythms are synchronized with seasons of the year and have a cycle of about 12 months. Examples include molting in periodical cicadas and migratory flights in monarch butterflies.

Insect Seasonal Cycle

- Looking at the cycles as they pass through different seasons of the year
- Very important to the survival of the insects
- Evolve their ways to time their activities to take advantage of environmental resources and avoid unfavorable extremes
- Important when attempting to reduce pest damage



- Seasonal cycles are grouped according to the no. of generations occur in a year → population Voltinity
- Voltinity types are;
 1. Univoltine
 2. Multivoltine
 3. Delayed voltine
- Univoltine → single generation each year (grasshoppers, some cassida beetles, rootworms)
- Multivoltine → more than one generations per year (house flies, thrips, aphids, etc.)
- Delayed voltine → take more than one year to complete one cycle (wire worms, June beetles)

Seasonal adaptations

- Adapts to many adverse environmental conditions
- Irregular unpredictable extreme
 - temperatures
 - moisture
 - food
 - crowding
 - other factors
- Usually localized and temporary
- Insect response to these stresses by becoming quiescent or moving out of the area

- Other adverse conditions occur regularly with the seasons
- Are predictable (freezing in temperate climate, drought in deserts, rains in tropics)
- Most insects avoid these condition by undergoing physiological and behavioural alterations
- Helps to identify patterns of their seasonal cycle

Dormancy

- Escaping regular seasonal adversities
- Growth, development and reproduction are suppressed
- May occur during summer, fall, winter or spring
- Some dormant through more than one seasons
- In temperate climates hibernation is the most prominent character

Diapause

- Dormancy, seasonal migration, and morphological changes are controlled by physiological condition known as diapause
- Low metabolism, little or no development, increase resistant to environmental extremes and altered behavioural activities
- Occur at specific genetically fixed stage of insect life cycle
- It continues for several weeks and may take longer too

THANK YOU!

